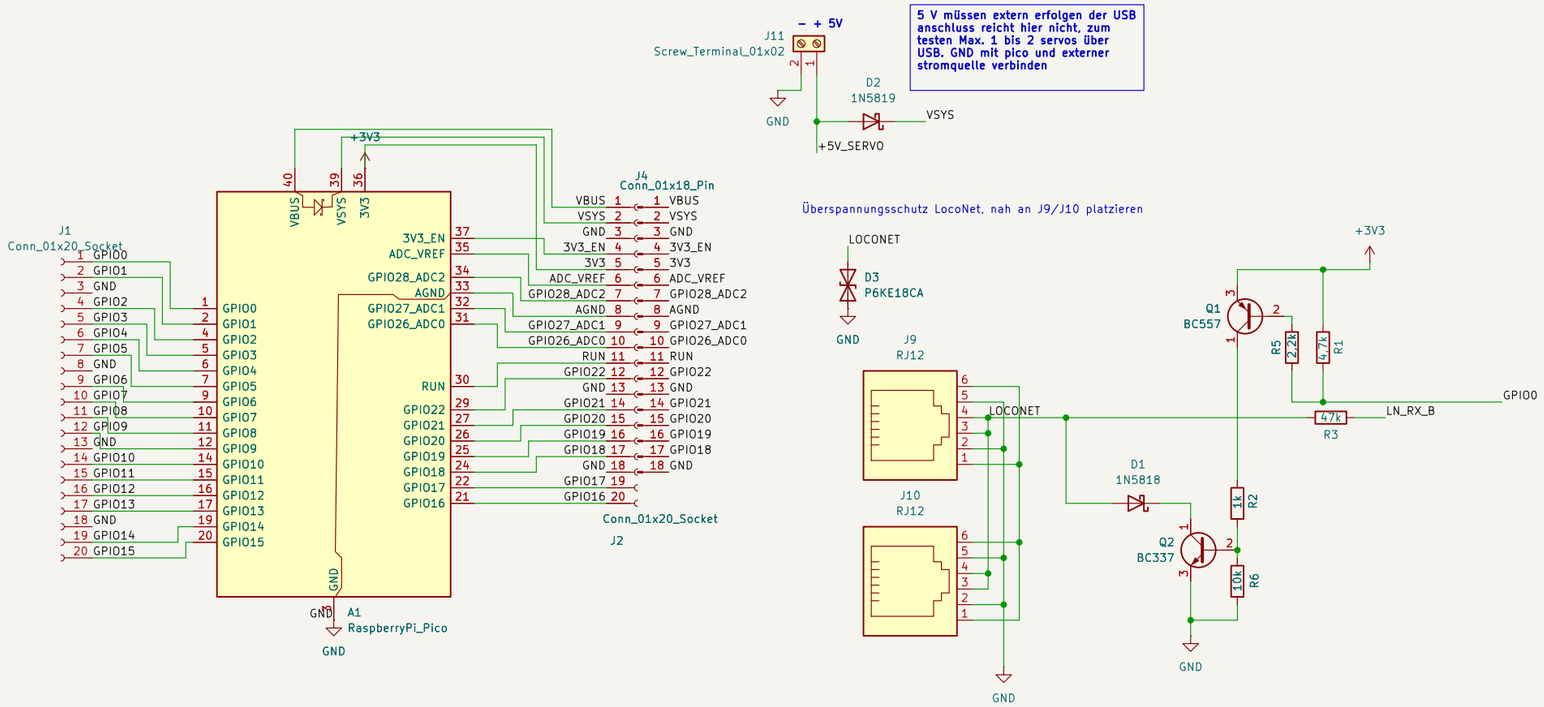

J11
Screw_Terminal_01x02
– + 5V
GND
D2
1N5819
VSYS
+5V_SERVO
5 V müssen extern erfolgen der USB anschluss reicht hier nicht, zum testen Max. 1 bis 2 servos über USB. GND mit pico und externer stromquelle verbinden
J1
Conn_01x20_Socket
1 GPIO0
2 GPIO1
3 GND
4 GPIO2
5 GPIO3
6 GPIO4
7 GPIO5
8 GND
9 GPIO6
10 GPIO7
11 GPIO8
12 GPIO9
13 GND
14 GPIO10
15 GPIO11
16 GPIO12
17 GPIO13
18 GND
19 GPIO14
20 GPIO15
+3V3
40 VBUS
39 VSYS
36 3V3
3V3_EN 37
ADC_VREF 35
GPIO28_ADC2 34
AGND 33
GPIO27_ADC1 32
GPIO26_ADC0 31
RUN 30
GPIO22 29
GPIO21 27
GPIO20 26
GPIO19 25
GPIO18 24
GPIO17 22
GPIO16 21
GND 38
GND 28
GND 23
GND 8
GND 18
GND 13
GND
A1
RaspberryPi_Pico
GND
J4
Conn_01x18_Pin
VBUS 1
VSYS 2
GND 3
3V3_EN 4
3V3 5
ADC_VREF 6
GPIO28_ADC2 7
AGND 8
GPIO27_ADC1 9
GPIO26_ADC0 10
RUN 11
GPIO22 12
GND 13
GPIO21 14
GPIO20 15
GPIO19 16
GPIO18 17
GND 18
GPIO17 19
GPIO16 20
Conn_01x20_Socket
J2
Überspannungsschutz LocoNet, nah an J9/J10 platzieren
LOCONET
D3
P6KE18CA
GND
J9
RJ12
J10
RJ12
LOCONET
GND
+3V3
Q1
BC557
R5
2.2k
4.7k
R1
GPIO0
47k
R3
LN_RX_B
D1
1N5818
1k
R2
Q2
BC337
10k
R6
GND
Puffer Servo–Versorgung, nah an J11 platzieren
+5V_SERVO
C1
1000µF/16V
C2
100nF
GND
GND
LocoNet–Eingang: Schwelle ca. 3,4–4 V (R3 47k / R7 10k),
Q4 invertiert zurück: Leitung High = RX High
+3V3
+3V3
10k
R8
10k
R10
GPIO1
10k
R9
Q4
BC337
Q3
BC337
LN_RX_B
10k
R7
GND
GND
GND
Servo–Anschlüsse: Pin 1 = Signal (GPIO), Pin 2 = +5V_SERVO, Pin 3 = GND
Gruppe 1: Servo 1-4
1 GPIO2
2 +5V_SERVO
3 GND
J12
Servo1
1 GPIO3
2 +5V_SERVO
3 GND
J13
Servo2
1 GPIO4
2 +5V_SERVO
3 GND
J14
Servo3
1 GPIO5
2 +5V_SERVO
3 GND
J15
Servo4
Gruppe 2: Servo 5-8
1 GPIO6
2 +5V_SERVO
3 GND
J16
Servo5
1 GPIO7
2 +5V_SERVO
3 GND
J17
Servo6
1 GPIO8
2 +5V_SERVO
3 GND
J18
Servo7
Gruppe 3: Servo 9-12
1 GPIO9
2 +5V_SERVO
3 GND
J19
Servo8
1 GPIO10
2 +5V_SERVO
3 GND
J20
Servo9
1 GPIO11
2 +5V_SERVO
3 GND
J21
Servo10
1 GPIO12
2 +5V_SERVO
3 GND
J22
Servo11
Gruppe 4: Servo 13-16
1 GPIO13
2 +5V_SERVO
3 GND
J23
Servo12
1 GPIO14
2 +5V_SERVO
3 GND
J24
Servo13
1 GPIO15
2 +5V_SERVO
3 GND
J25
Servo14
1 GPIO16
2 +5V_SERVO
3 GND
J26
Servo15
1 GPIO17
2 +5V_SERVO
3 GND
J27
Servo16
Sheet: /
File: loconet–pico.kicad_sch
Title: Loconet mit 16 Servos und Raspi Pico
Size: A3
Date: 12.11.2026
Rev:
KiCad E.D.A. 10.0.7
Id: 1/1